

31. Compare Match Timer (CMT)

This MCU has two on-chip compare match timer (CMT) units (unit 0 and unit 1), each consisting of a two-channel 16-bit timer (i.e., a total of four channels). The CMT has a 16-bit counter, and can generate interrupts at set intervals.

In this section, “PCLK” is used to refer to PCLKB.

31.1 Overview

Table 31.1 lists the specifications for the CMT.

Figure 31.1 shows a block diagram of the CMT (unit 0). A two-channel CMT constitutes a unit. Unit 0 and unit 1 are the same in terms of specifications. Compare match timer start register 0 (CMSTR0) and compare match interrupts (CMI0 and CMI1) of unit 0 correspond to compare match timer start register 1 (CMSTR1) and compare match interrupts (CMI2 and CMI3) of unit 1.

Table 31.1 CMT Specifications

Item	Description
Count clocks	Four frequency dividing clocks One clock from PCLK/8, PCLK/32, PCLK/128, and PCLK/512 can be selected for each channel.
Interrupt	A compare match interrupt can be requested for each channel.
Event link function (output)	An event signal is output upon a CMT1 compare match.
Event link function (input)	Linking to the specified module is possible. CMT1 count start, event counter, or count restart operation is possible.
Low power consumption function	Each unit can be placed in a module stop state.

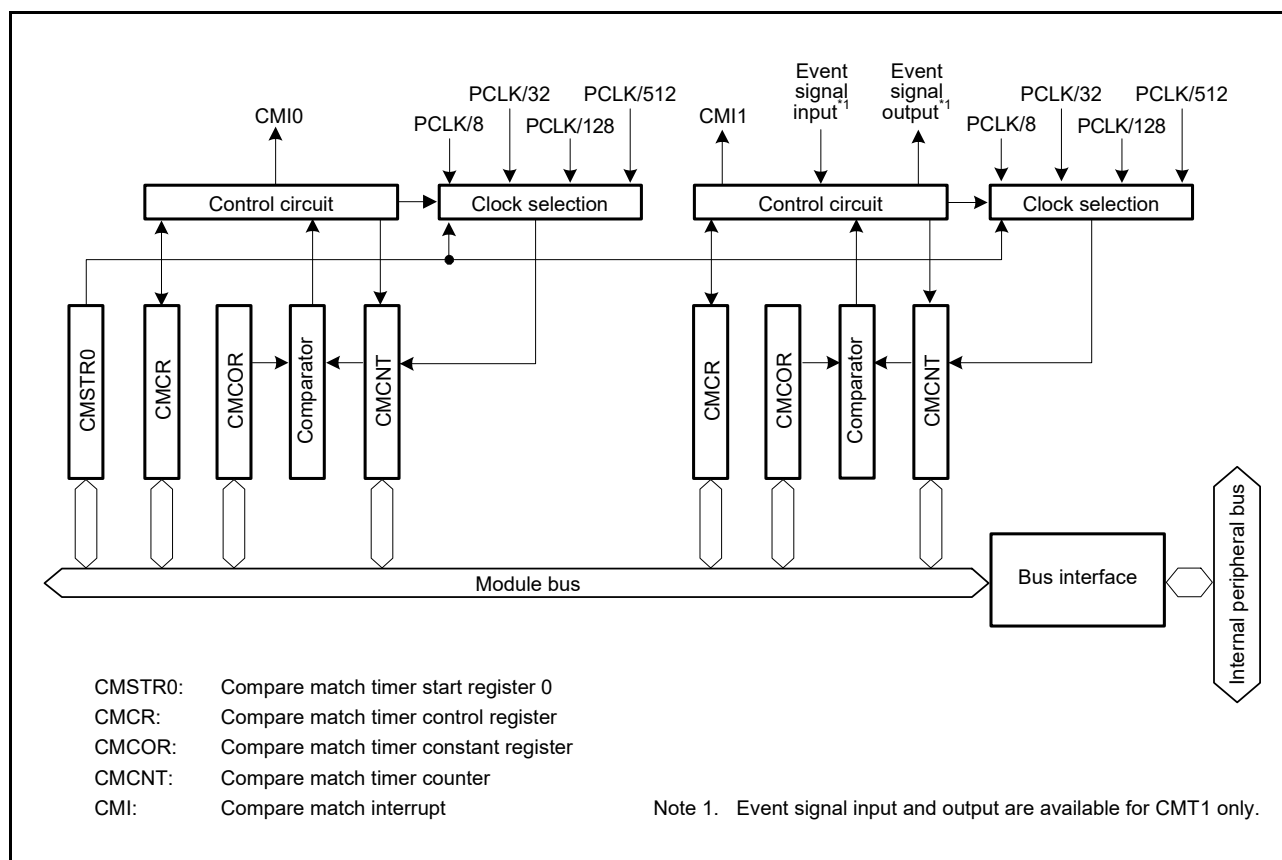


Figure 31.1 CMT (Unit 0) Block Diagram

31.2 Register Descriptions

31.2.1 Compare Match Timer Start Register 0 (CMSTR0)

Address(es): 0008 8000h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STR1	STR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	STR0	Count Start 0	0: CMT0.CMCNT count is stopped. 1: CMT0.CMCNT count is started.	R/W
b1	STR1	Count Start 1	0: CMT1.CMCNT count is stopped. 1: CMT1.CMCNT count is started.	R/W
b15 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

31.2.2 Compare Match Timer Start Register 1 (CMSTR1)

Address(es): 0008 8010h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STR3	STR2
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	STR2	Count Start 2	0: CMT2.CMCNT count is stopped. 1: CMT2.CMCNT count is started.	R/W
b1	STR3	Count Start 3	0: CMT3.CMCNT count is stopped. 1: CMT3.CMCNT count is started.	R/W
b15 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

31.2.3 Compare Match Timer Control Register (CMCR)

Address(es): CMT0.CMCR 0008 8002h, CMT1.CMCR 0008 8008h, CMT2.CMCR 0008 8012h, CMT3.CMCR 0008 8018h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	CMIE	—	—	—	—	CKS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKS[1:0]	Clock Select	b1 b0 0 0: PCLK/8 0 1: PCLK/32 1 0: PCLK/128 1 1: PCLK/512	R/W
b5 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	CMIE	Compare Match Interrupt Enable	0: Compare match interrupt (CMIn) disabled 1: Compare match interrupt (CMIn) enabled	R/W
b7	—	Reserved	This bit is read as undefined. The write value should be 1.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

CKS[1:0] Bits (Clock Select)

These bits select the count source from four frequency dividing clocks obtained by dividing the peripheral module clock (PCLK).

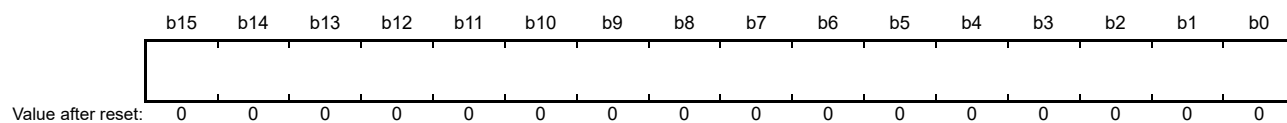
When the CMSTRm.STRn (m = 0, 1; n = 0 to 3) bit is set to 1, the CMCNT counter starts counting up on the clock selected with the CKS[1:0] bits.

CMIE Bit (Compare Match Interrupt Enable)

The CMIE bit enables or disables compare match interrupt (CMIn) (n = 0 to 3) generation when the CMCNT counter and the CMCOR register values match.

31.2.4 Compare Match Counter (CMCNT)

Address(es): CMT0.CMCNT 0008 8004h, CMT1.CMCNT 0008 800Ah, CMT2.CMCNT 0008 8014h, CMT3.CMCNT 0008 801Ah



The CMCNT counter is a readable/writable up-counter.

When an frequency dividing clock is selected by the CMCR.CKS[1:0] bits and the CMSTRm.STRn (m = 0, 1; n = 0 to 3) bit is set to 1, the CMCNT counter starts counting up using the selected clock.

When the value in the CMCNT counter and the value in the CMCOR register match, the CMCNT counter is set to 0000h. At the same time, a compare match interrupt (CMIn) (n = 0 to 3) is generated.

31.2.5 Compare Match Constant Register (CMCOR)

Address(es): CMT0.CMCOR 0008 8006h, CMT1.CMCOR 0008 800Ch, CMT2.CMCOR 0008 8016h, CMT3.CMCOR 0008 801Ch



The CMCOR register is a readable/writable register to set a value for compare match with the CMCNT counter.

31.3 Operation

31.3.1 Periodic Count Operation

When an frequency dividing clock is selected by the CMCR.CKS[1:0] bits and the CMSTRm.STRn ($m = 0, 1$; $n = 0$ to 3) bit is set to 1, the CMCNT counter starts counting up using the selected clock.

When the value in the counter and the value in the register match, a compare match interrupt (CMIn) ($n = 0$ to 3) is generated. The CMCNT counter then starts counting up again from 0000h. Figure 31.2 shows the operation of the CMCNT counter.

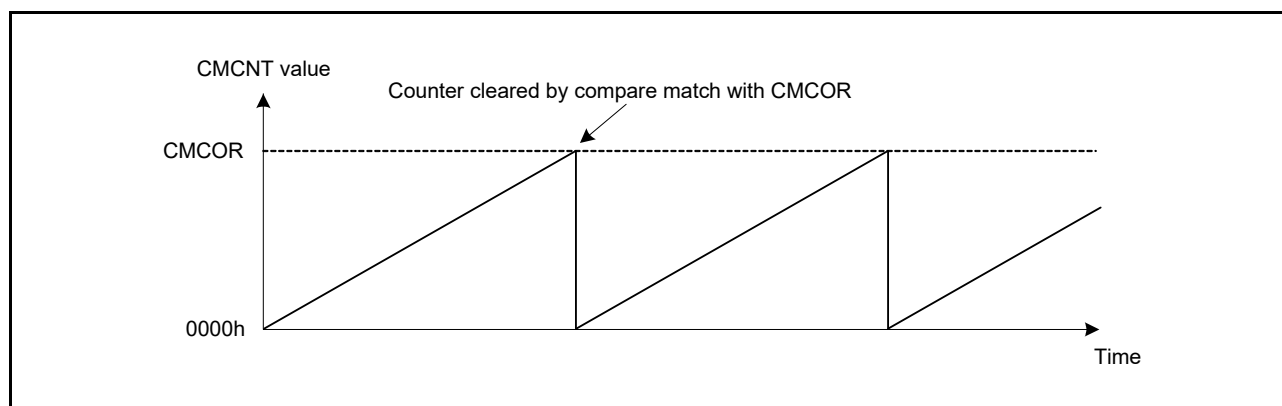


Figure 31.2 CMCNT Counter Operation

31.3.2 CMCNT Count Timing

As the count clock to be input to the CMCNT counter, one of four frequency dividing clocks (PCLK/8, PCLK/32, PCLK/128, and PCLK/512) obtained by dividing the peripheral module clock (PCLK) can be selected with the CMCR.CKS[1:0] bits. Figure 31.3 shows the timing of the CMCNT counter.

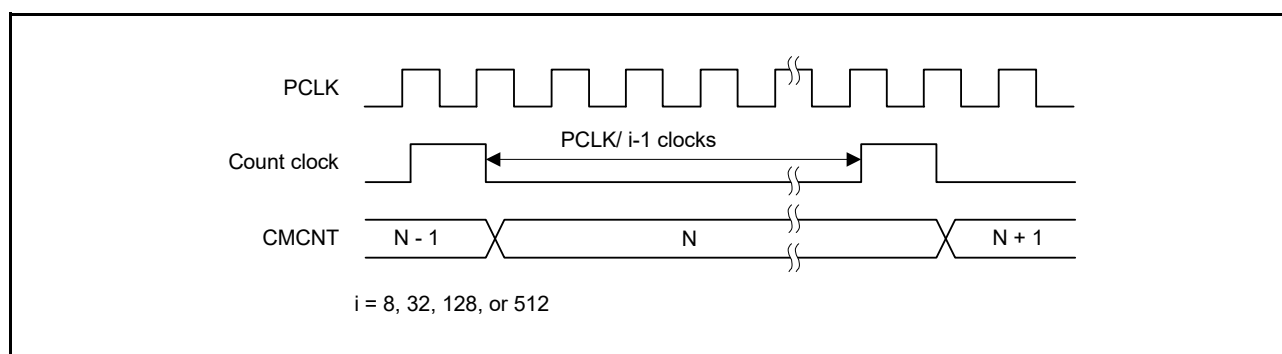


Figure 31.3 CMCNT Count Timing

31.4 Interrupts

31.4.1 Interrupt Sources

The CMT has channels and each of them to which a different vector address is allocated has a compare match interrupt (CMI_n) ($n = 0$ to 3). When a compare match interrupt occurs, the corresponding interrupt request is output.

When the interrupt request is used to generate a CPU interrupt, the priority of channels can be changed by the interrupt controller settings. For details, see section 15, Interrupt Controller (ICUD).

Table 31.2 CMT Interrupt Sources

Name	Interrupt Sources	DTC Activation	DMAC Activation
CMI0	Compare match in CMT0	Possible	Possible
CMI1	Compare match in CMT1	Possible	Possible
CMI2	Compare match in CMT2	Possible	Possible
CMI3	Compare match in CMT3	Possible	Possible

31.4.2 Timing of Compare Match Interrupt Generation

When the CMCNT counter and the CMCOR register match, a compare match interrupt (CMI_n) ($n = 0$ to 3) is generated. A compare match signal is generated at the last state in which the values match (the timing when the CMCNT counter updates the matched count value). That is, after a match between the CMCOR register and the CMCNT counter, the compare match signal is not generated until the next the CMCNT counter input clock.

Figure 31.4 shows the timing of a compare match interrupt.

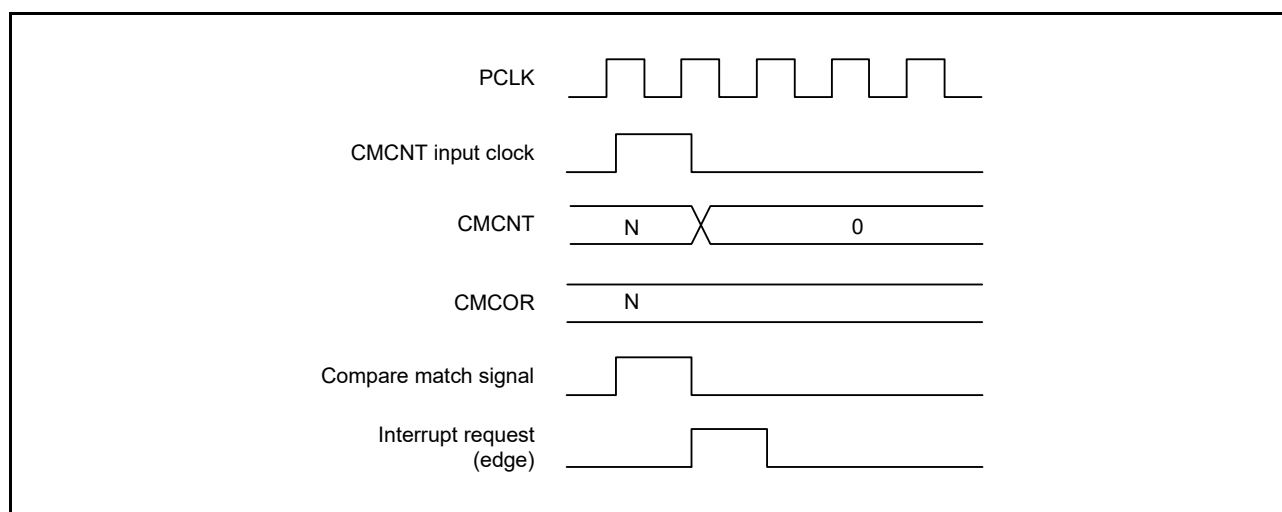


Figure 31.4 Timing of a Compare Match Interrupt

31.5 Link Operations by ELC

31.5.1 Event Signal Output to ELC

The CMT uses the event link controller (ELC) to perform link operation to a preset module using the interrupt request signal as the event signal. The CMT outputs the event signal upon a CMT1 compare match.

The event signal can be output regardless of the setting of the corresponding interrupt request enable bit (CMTn.CMCR.CMIE).

31.5.2 CMT Operation When Receiving an Event Signal from ELC

The CMT can perform either of the following operations upon the event preset by the ELSR7 register of the ELC.

(1) Count Start

When the CMT count start operation is selected by the ELOPC register of the ELC and the event specified by ELSR7 occurs, the CMSTR0.STR1 bit is set to 1, starting the CMT count operation.

However, if the specified event occurs while the CMSTR0.STR1 bit is 1, the event is ignored.

(2) Event Count

When the CMT event count operation is selected by the ELOPC register of the ELC and the event specified by ELSR7 occurs with the CMSTR0.STR1 bit being 1, the events are counted as the count source regardless of the CMT1.CMCR.CKS[1:0] bit setting. Reading the counter value returns the number of events that have been actually input.

(3) Count Restart

When the CMT count restart operation is selected by the ELOPC register of the ELC and the event specified by ELSR7 occurs, the CMT1.CMCNT counter value is modified to the initial value. If the CMSTR0.STR1 bit is 1 here, the count operation can be continued.

31.5.3 Notes on Operating CMT According to an Event Signal from ELC

The following describes the notes on operating the CMT using the event link feature.

(1) Count Start

When the event specified by ELSR7 occurs during the write cycle to the CMSTR0.STR1 bit, the cycle is not completed; setting 1 according to the event occurrence takes priority.

(2) Event Count

When the event specified by ELSR7 occurs during the write cycle to the CMT1.CMCNT, the cycle is not completed; event count operation according to the event occurrence takes priority.

(3) Count Restart

When the event specified by ELSR7 occurs during the write cycle to the CMT1.CMCNT, the cycle is not completed; count value initialization according to the event occurrence takes priority.

31.6 Usage Notes

31.6.1 Setting the Module Stop Function

The CMT can be enabled or disabled using the module stop control register. After a reset, the CMT is in the module stop state. The registers can be accessed by releasing the module stop state. For details, refer to [section 11, Low Power Consumption](#).

31.6.2 Conflict between CMCNT Counter Writing and Compare Match

When the compare match signal is generated while writing to the CMCNT counter, clearing the CMCNT counter has priority over writing to it. In this case, the CMCNT counter is not written to. [Figure 31.5](#) shows the timing to clear the CMCNT counter.

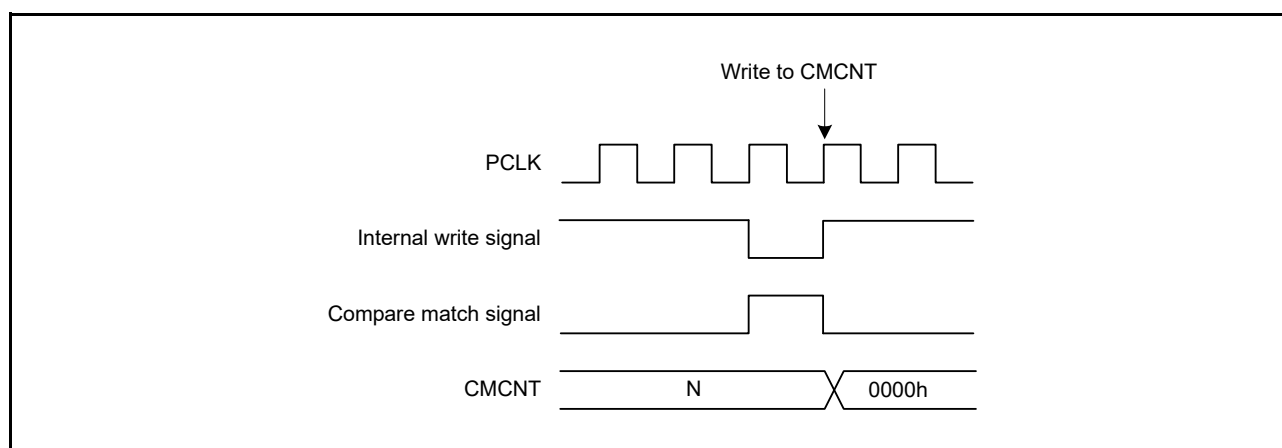


Figure 31.5 Conflict between CMCNT Counter Writing and Compare Match

31.6.3 Conflict between CMCNT Counter Writing and Incrementing

If writing to the counter and the incrementing conflict, the writing has priority over the incrementing. [Figure 31.6](#) shows the timing to write the CMCNT counter.

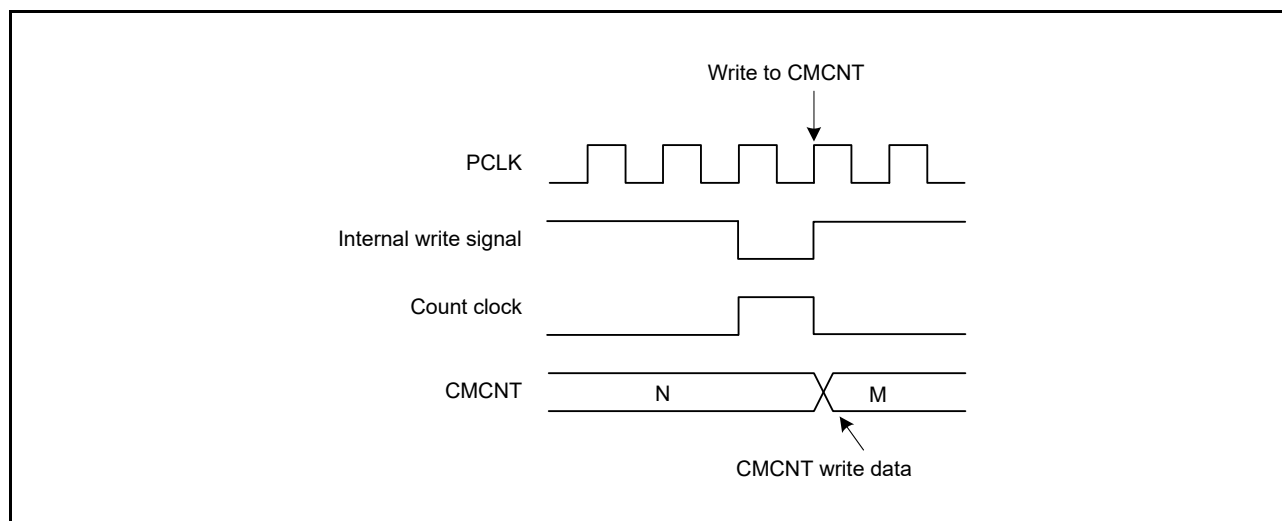


Figure 31.6 Conflict between CMCNT Counter Writing and Incrementing